

## **Sid Mathur's Thematic ETF Screener – Relative-Strength Rotation**

2026-08-27

|                           |    |
|---------------------------|----|
| <b>Contents</b>           |    |
| Action Center             | 3  |
| Performance vs benchmarks | 5  |
| Position Sheet            | 9  |
| Sector Overview           | 10 |
| Leading Cluster Details   | 11 |
| Closed Trades             | 22 |
| ETF RS Leaders & Laggards | 23 |
| Watchlist                 | 25 |
| Run Parameters            | 26 |

## Action Center

**As of 27-Aug-2026:** holding **18 ETFs** across **6 sectors** (of **8** currently in relative-LONG). Gross exposure **100%**, cash **0%**.

Full-period: **CAGR 12.3%**, **Sharpe 0.87**, **Sortino 1.22**, **Max DD -14.3%**, Calmar **0.86**. Signal on relative strength, P&L on the absolute ETFs held; no leverage; inverse-vol weighted within the selected leader set.

Rebalance is **monthly** (last trading day decision, next-morning execution). Signals are computed daily, but positions only change at month-end; intra-month the book is static. Min-hold 30 days is honored on each per-position basis.

## Latest rebal changes

Last rebalance: **03-Aug-2026** (29 trades).

## Entered

| ticker | name                             | sector                 | weight |
|--------|----------------------------------|------------------------|--------|
| EWS    | iShares MSCI Singapore ETF       | Developed ex-US Equity | 5.5%   |
| EUFN   | iShares MSCI Europe Financial... | Developed ex-US Equity | 3.8%   |
| PWV    | Invesco Large Cap Value ETF      | U.S. Dividend Equity   | 8.1%   |
| IWD    | iShares Russell 1000 Value ETF   | U.S. Large Cap Value   | 7.2%   |
| MGV    | Vanguard Mega Cap Value ETF      | U.S. Large Cap Value   | 7.8%   |
| PEY    | Invesco High Yield Equity Div... | U.S. Dividend Equity   | 5.4%   |
| ITA    | iShares U.S. Aerospace & Defe... | U.S. Large Cap Equity  | 3.3%   |
| RDIV   | Invesco S&P Ultra Dividend Re... | U.S. Dividend Equity   | 5.9%   |
| IWS    | iShares Russell Mid-Cap Value... | U.S. Large Cap Value   | 6.6%   |
| QDF    | FlexShares Quality Dividend I... | U.S. Large Cap Equity  | 7.0%   |
| QUAL   | iShares MSCI USA Quality Fact... | U.S. Large Cap Equity  | 7.2%   |
| RWR    | State Street SPDR Dow Jones R... | Real Estate            | 5.1%   |
| EPOL   | iShares MSCI Poland ETF          | Developed ex-US Equity | 3.4%   |
| ETHO   | Amplify Etho Climate Leadersh... | U.S. Mid Cap Equity    | 5.6%   |
| RWJ    | Invesco S&P SmallCap 600 Reve... | U.S. Mid Cap Equity    | 5.4%   |
| INDS   | Pacer Industrial Real Estate ... | Real Estate            | 5.0%   |
| PSR    | Invesco Active U.S. Real Esta... | Real Estate            | 5.2%   |

## Exited

| ticker | name                             | sector                  | prev_wt |
|--------|----------------------------------|-------------------------|---------|
| ARKG   | ARK Genomic Revolution ETF       | Healthcare              | 4.9%    |
| XBI    | State Street SPDR S&P Biotech... | Healthcare              | 7.7%    |
| AAXJ   | iShares MSCI All Country Asia... | Emerging Markets Equity | 6.5%    |
| EWT    | iShares MSCI Taiwan ETF          | Emerging Markets Equity | 5.6%    |
| AIA    | iShares Asia 50 ETF              | Emerging Markets Equity | 5.2%    |
| PTH    | Invesco Dorsey Wright Healthc... | Healthcare              | 8.0%    |
| RDVY   | First Trust Rising Dividend A... | U.S. Large Cap Value    | 14.2%   |
| VIS    | Vanguard Industrials ETF         | U.S. Large Cap Value    | 10.6%   |
| IWC    | iShares Microcap ETF             | U.S. Mid Cap Equity     | 8.8%    |
| IJT    | iShares S&P SmallCap 600 Grow... | U.S. Mid Cap Equity     | 13.1%   |
| PAVE   | Global X U.S. Infrastructure ... | U.S. Large Cap Value    | 9.4%    |

## Imminent moves

Next-in-line entries (eligible, in a leading sector, not yet held)

| ticker | name                             | sector                  | rs_momentum |
|--------|----------------------------------|-------------------------|-------------|
| GDX    | VanEck Gold Miners ETF           | Gold                    | 13.0%       |
| GDXJ   | VanEck Junior Gold Miners ETF    | Gold                    | 11.0%       |
| SIL    | Global X Silver Miners ETF       | Gold                    | 6.8%        |
| NANR   | State Street SPDR S&P North A... | Materials               | 6.4%        |
| COPX   | Global X Copper Miners ETF       | Materials               | 5.6%        |
| EWQ    | iShares MSCI Austria ETF         | Developed ex-US Equity  | 5.0%        |
| SILJ   | Amplify Junior Silver Miners ... | Gold                    | 3.9%        |
| EWT    | iShares MSCI Taiwan ETF          | Emerging Markets Equity | 3.4%        |
| EWG    | iShares MSCI Canada Index Fund   | Developed ex-US Equity  | 3.3%        |
| GNR    | State Street SPDR S&P Global ... | Materials               | 3.1%        |

## Performance vs benchmarks

Table 4: Same window. Defensive Strategy = dual-momentum clusters; Market = full-universe PC1 buy-and-hold.

| Metric   | Rel-Strength | Defensive Strategy | Market |
|----------|--------------|--------------------|--------|
| CAGR     | 12.3%        | 5.4%               | 7.5%   |
| Ann vol  | 13.5%        | 8.2%               | 14.9%  |
| Sharpe   | 0.87         | 0.65               | 0.56   |
| Sortino  | 1.22         | 0.91               | 0.81   |
| Max DD   | -14.3%       | -13.1%             | -26.1% |
| Calmar   | 0.86         | 0.42               | 0.29   |
| Hit rate | 55%          | 54%                | 53%    |

Table 5: Year-by-year.

| year | start_nav | end_nav | return | sharpe | max_drawdown | n_obs |
|------|-----------|---------|--------|--------|--------------|-------|
| 2021 | 100.0     | 100.0   | 0.0%   |        | 0.0%         | 74    |
| 2022 | 100.0     | 97.4    | -2.6%  | -0.60  | -3.9%        | 251   |
| 2023 | 97.3      | 95.1    | -2.4%  | -0.21  | -9.7%        | 250   |
| 2024 | 93.4      | 107.6   | 15.2%  | 0.93   | -10.9%       | 252   |
| 2025 | 106.9     | 142.5   | 33.3%  | 2.02   | -9.7%        | 250   |
| 2026 | 145.1     | 176.9   | 21.9%  | 1.42   | -14.3%       | 163   |

## Monthly returns

Table 6: Monthly returns by year (live window).

|      | Jan   | Feb   | Mar    | Apr   | May   | Jun  | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------|-------|-------|--------|-------|-------|------|-------|-------|-------|-------|-------|-------|
| 2021 |       |       |        |       |       |      |       |       | 0.0%  | 0.0%  | 0.0%  | 0.0%  |
| 2022 | 0.0%  | 0.0%  | 0.0%   | 0.0%  | 0.0%  | 0.0% | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0%  | -2.6% |
| 2023 | -0.1% | -3.6% | -0.4%  | 1.9%  | -1.9% | 0.3% | 4.6%  | -6.2% | -1.1% | -1.1% | -0.1% | 5.7%  |
| 2024 | -2.9% | 6.4%  | 3.7%   | -5.1% | 3.8%  | 1.7% | 1.1%  | 0.0%  | 2.7%  | -0.4% | 7.9%  | -5.5% |
| 2025 | 3.8%  | 3.2%  | 0.5%   | -0.1% | 0.0%  | 7.8% | 1.5%  | 3.1%  | 5.2%  | 2.8%  | -2.6% | 3.6%  |
| 2026 | 8.4%  | 7.5%  | -10.0% | 12.4% | 7.4%  | 1.6% | -5.1% | 1.6%  |       |       |       |       |

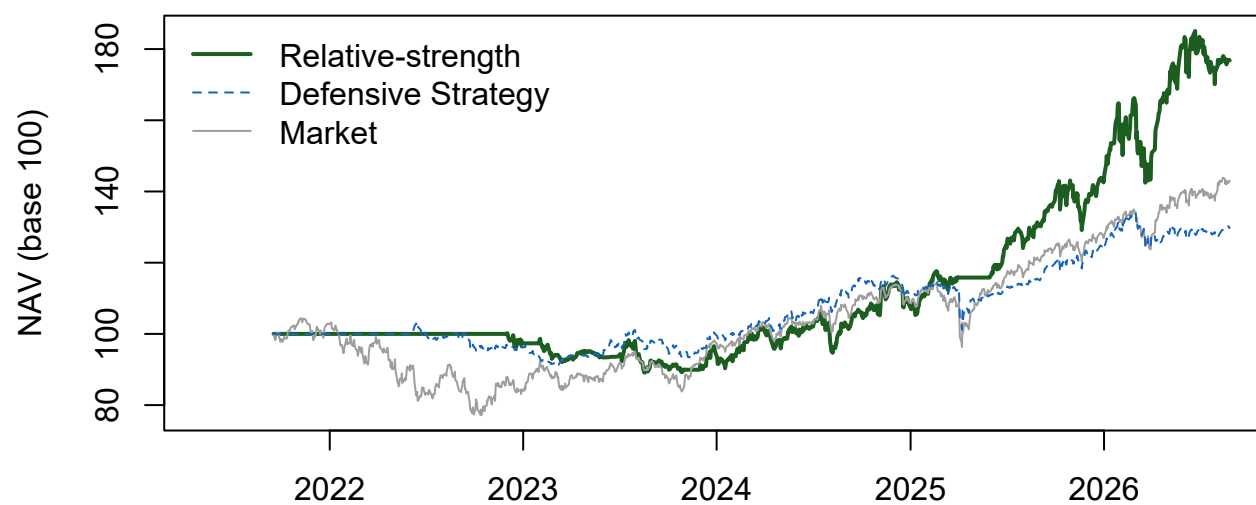


Figure 1: Growth of 100. Relative-strength rotation vs Defensive Strategy and market buy-and-hold.

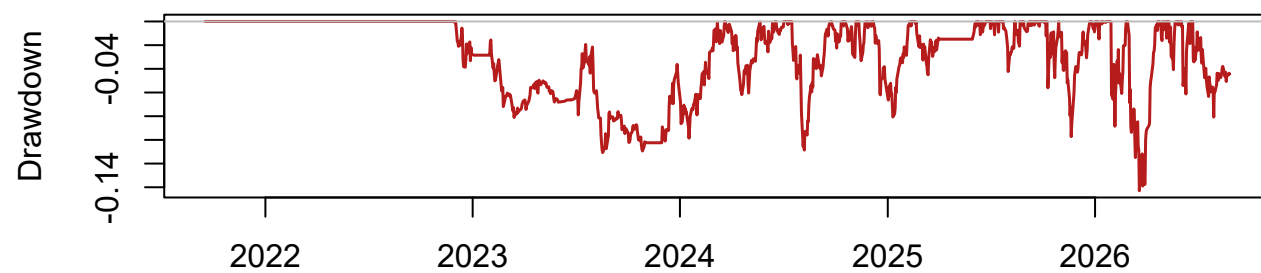


Figure 2: Drawdown of the relative-strength strategy (live window).

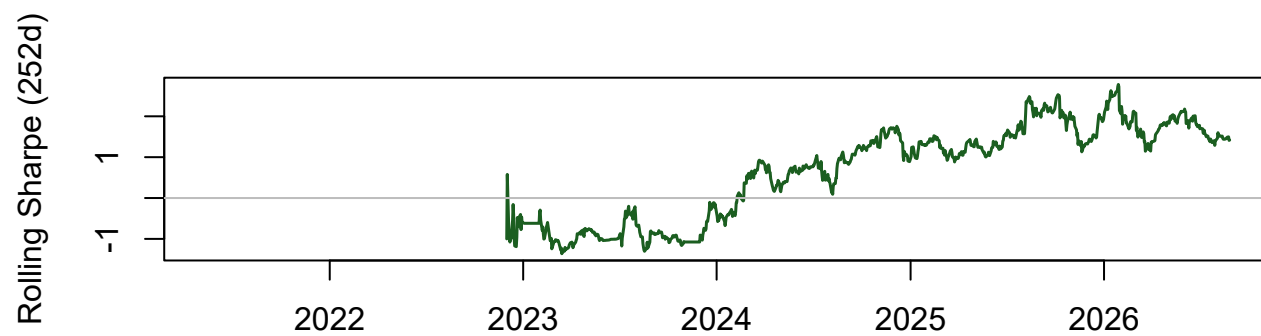


Figure 3: Rolling 252-day annualised Sharpe.



## Position Sheet

Current holdings, enriched with per-name vol, recent return, and the per-name signal state. Sorted by sector then weight desc.

Table 7: Current book. Vol-60d = trailing-60d annualised; ret-3m = 63-day cumulative; days held = calendar days in book.

| ticker | name                           | sector                 | weight | vol_60d | ret_3m | days_held | etf_rs | sector_rs |
|--------|--------------------------------|------------------------|--------|---------|--------|-----------|--------|-----------|
| EWS    | iShares MSCI Singapore ETF     | Developed ex-US Equity | 5.5%   | 16.0%   | 15.7%  | 23        | LONG   | LONG      |
| EUFN   | iShares MSCI Europe Financi... | Developed ex-US Equity | 3.8%   | 18.5%   | 10.7%  | 23        | FLAT   | LONG      |
| EPOL   | iShares MSCI Poland ETF        | Developed ex-US Equity | 3.4%   | 21.8%   | 9.5%   | 23        | LONG   | LONG      |
| PSR    | Invesco Active U.S. Real Es... | Real Estate            | 5.2%   | 16.2%   | 0.9%   | 23        | LONG   | LONG      |
| RWR    | State Street SPDR Dow Jones... | Real Estate            | 5.1%   | 15.7%   | 2.9%   | 23        | FLAT   | LONG      |
| INDS   | Pacer Industrial Real Estat... | Real Estate            | 5.0%   | 14.1%   | 1.7%   | 23        | FLAT   | LONG      |
| PWV    | Invesco Large Cap Value ETF    | U.S. Dividend Equity   | 8.1%   | 10.2%   | 11.4%  | 23        | LONG   | LONG      |
| RDIV   | Invesco S&P Ultra Dividend ... | U.S. Dividend Equity   | 5.9%   | 14.0%   | 8.8%   | 23        | FLAT   | LONG      |
| PEY    | Invesco High Yield Equity D... | U.S. Dividend Equity   | 5.4%   | 15.8%   | 11.2%  | 23        | LONG   | LONG      |
| QUAL   | iShares MSCI USA Quality Fa... | U.S. Large Cap Equity  | 7.2%   | 11.6%   | 3.5%   | 23        | FLAT   | FLAT      |
| QDF    | FlexShares Quality Dividend... | U.S. Large Cap Equity  | 7.0%   | 12.0%   | 3.7%   | 23        | FLAT   | FLAT      |
| ITA    | iShares U.S. Aerospace & De... | U.S. Large Cap Equity  | 3.3%   | 25.0%   | -0.4%  | 23        | FLAT   | FLAT      |
| MGV    | Vanguard Mega Cap Value ETF    | U.S. Large Cap Value   | 7.8%   | 10.7%   | 6.7%   | 23        | LONG   | FLAT      |
| IWD    | iShares Russell 1000 Value ... | U.S. Large Cap Value   | 7.2%   | 11.3%   | 8.4%   | 23        | LONG   | FLAT      |
| IWS    | iShares Russell Mid-Cap Val... | U.S. Large Cap Value   | 6.6%   | 12.4%   | 7.2%   | 23        | FLAT   | FLAT      |
| ETHO   | Amplify Etho Climate Leader... | U.S. Mid Cap Equity    | 5.6%   | 15.6%   | 7.0%   | 23        | LONG   | FLAT      |
| RWJ    | Invesco S&P SmallCap 600 Re... | U.S. Mid Cap Equity    | 5.4%   | 15.0%   | 7.9%   | 23        | FLAT   | FLAT      |
| JETS   | U.S. Global Jets ETF           | U.S. Mid Cap Equity    | 2.3%   | 33.6%   | 0.8%   | 56        | LONG   | FLAT      |

## Sector Overview

Table 8: All clusters, LONG first then FLAT; within each, RS-momentum desc. Held = positions taken; Members = ETFs in the cluster.

| sector                           | rs_state | rs_momentum | n_held | n_members |
|----------------------------------|----------|-------------|--------|-----------|
| U.S. Dividend Equity             | LONG     | 3.7%        | 3      | 25        |
| Gold                             | LONG     | 1.5%        | 0      | 12        |
| Developed ex-US Equity           | LONG     | 0.5%        | 3      | 79        |
| Materials                        | LONG     | 0.5%        | 0      | 17        |
| Japan Equity                     | LONG     | 0.3%        | 0      | 9         |
| Real Estate                      | LONG     | -1.1%       | 3      | 14        |
| Broad Commodities                | LONG     | -1.6%       | 0      | 9         |
| Emerging Markets Equity          | LONG     | -3.6%       | 0      | 30        |
| Healthcare                       | FLAT     | 16.9%       | 0      | 17        |
| Financials                       | FLAT     | 7.8%        | 0      | 14        |
| Technology                       | FLAT     | 4.6%        | 0      | 14        |
| U.S. Large Cap Value             | FLAT     | 2.5%        | 3      | 60        |
| U.S. Mid Cap Equity              | FLAT     | 1.5%        | 3      | 72        |
| Consumer Staples                 | FLAT     | -2.4%       | 0      | 11        |
| U.S. Large Cap Equity            | FLAT     | -2.5%       | 3      | 81        |
| Cash / Ultra-Short Fixed Income  | FLAT     | -2.8%       | 0      | 9         |
| Bank Loan / Floating Rate        | FLAT     | -3.0%       | 0      | 9         |
| International Fixed Income       | FLAT     | -3.1%       | 0      | 6         |
| High Yield Bond                  | FLAT     | -3.6%       | 0      | 23        |
| Aggregate Fixed Income           | FLAT     | -3.8%       | 0      | 58        |
| Municipal Bond                   | FLAT     | -4.0%       | 0      | 13        |
| Leveraged / Inverse              | FLAT     | -5.1%       | 0      | 2         |
| India Equity                     | FLAT     | -5.2%       | 0      | 12        |
| Inflation-Protected Fixed Income | FLAT     | -5.5%       | 0      | 10        |
| China Equity                     | FLAT     | -6.1%       | 0      | 15        |
| U.S. Large Cap Growth            | FLAT     | -6.3%       | 0      | 38        |

Leading Cluster Details

Sector 8: U.S. Dividend Equity

- Relative-strength state: **LONG** RS momentum (63d): **3.7%** held: **3 / 25** members
- Theme: U.S. Dividend Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| PWV    | Invesco Large Cap Value ETF    | 8.1%   | 4.1%   | LONG   | 23        |
| RDIV   | Invesco S&P Ultra Dividend ... | 5.9%   | 2.4%   | FLAT   | 23        |
| PEY    | Invesco High Yield Equity D... | 5.4%   | 5.2%   | LONG   | 23        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| SCHD   | Schwab US Dividend Equity E... | 89.1%  |          |

Sector 17: Gold

- Relative-strength state: **LONG** RS momentum (63d): **1.5%** held: **0 / 12** members
- Theme: Gold



No member of this sector currently passes both the relative-LONG and the absolute overlay.

Strategy-1 replication basket (context):

| ticker | name                      | rep_wt | held_now |
|--------|---------------------------|--------|----------|
| GLD    | SPDR Gold Shares          | 2.4%   |          |
| IAU    | iShares Gold Trust Shares | 75.4%  |          |
| SLV    | iShares Silver Trust      | 41.1%  |          |

Sector 11: Materials

- Relative-strength state: **LONG** RS momentum (63d): **0.5%** held: **0 / 17** members
- Theme: Materials



No member of this sector currently passes both the relative-LONG and the absolute overlay.

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| AMLP   | Alerian MLP ETF                | 13.9%  |          |
| URA    | Global X Uranium ETF           | 9.6%   |          |
| GUNR   | FlexShares Global Upstream ... | 77.2%  |          |

Sector 2: Developed ex-US Equity

- Relative-strength state: **LONG** RS momentum (63d): **0.5%** held: **3 / 79** members
- Theme: Developed ex-US Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| EWS    | iShares MSCI Singapore ETF     | 5.5%   | 12.6%  | LONG   | 23        |
| EUFN   | iShares MSCI Europe Financi... | 3.8%   | 6.2%   | FLAT   | 23        |
| EPOL   | iShares MSCI Poland ETF        | 3.4%   | 6.9%   | LONG   | 23        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| VEA    | Vanguard FTSE Developed Mar... | 91.1%  |          |

Sector 21: Japan Equity

- Relative-strength state: **LONG** RS momentum (63d): **0.3%** held: **0 / 9** members
- Theme: Japan Equity



No member of this sector currently passes both the relative-LONG and the absolute overlay.

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| EWJ    | iShares MSCI Japan Index Fu... | 90.7%  |          |

Sector 14: Real Estate

- Relative-strength state: **LONG** RS momentum (63d): **-1.1%** held: **3 / 14** members
- Theme: Real Estate



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| PSR    | Invesco Active U.S. Real Es... | 5.2%   | 0.8%   | LONG   | 23        |
| RWR    | State Street SPDR Dow Jones... | 5.1%   | 0.7%   | FLAT   | 23        |
| INDS   | Pacer Industrial Real Estat... | 5.0%   | -0.1%  | FLAT   | 23        |

Strategy-1 replication basket (context):

| ticker | name                     | rep_wt | held_now |
|--------|--------------------------|--------|----------|
| VNQ    | Vanguard Real Estate ETF | 96.8%  |          |



Sector 22: Broad Commodities

- Relative-strength state: **LONG** RS momentum (63d): **-1.6%** held: **0 / 9** members
- Theme: Broad Commodities



No member of this sector currently passes both the relative-LONG and the absolute overlay.

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| PDBC   | Invesco Optimum Yield Diver... | 118.1% |          |

Sector 7: Emerging Markets Equity

- Relative-strength state: **LONG** RS momentum (63d): **-3.6%** held: **0 / 30** members
- Theme: Emerging Markets Equity



No member of this sector currently passes both the relative-LONG and the absolute overlay.

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| IEMG   | iShares Core MSCI Emerging ... | 90.8%  |          |

Sector 3: U.S. Mid Cap Equity

- Relative-strength state: **FLAT** RS momentum (63d): **1.5%** held: **3 / 72** members
- Theme: U.S. Mid Cap Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| ETHO   | Amplify Etho Climate Leader... | 5.6%   | 3.3%   | LONG   | 23        |
| RWJ    | Invesco S&P SmallCap 600 Re... | 5.4%   | 4.1%   | FLAT   | 23        |
| JETS   | U.S. Global Jets ETF           | 2.3%   | -2.6%  | LONG   | 56        |

Strategy-1 replication basket (context):

| ticker | name                         | rep_wt | held_now |
|--------|------------------------------|--------|----------|
| IJH    | iShares Core S&P Mid-Cap ETF | 101.5% |          |

Sector 4: U.S. Large Cap Value

- Relative-strength state: **FLAT** RS momentum (63d): **2.5%** held: **3 / 60** members
- Theme: U.S. Large Cap Value



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| MGV    | Vanguard Mega Cap Value ETF    | 7.8%   | 1.3%   | LONG   | 23        |
| IWD    | iShares Russell 1000 Value ... | 7.2%   | 3.1%   | LONG   | 23        |
| IWS    | iShares Russell Mid-Cap Val... | 6.6%   | 2.3%   | FLAT   | 23        |

Strategy-1 replication basket (context):

| ticker | name               | rep_wt | held_now |
|--------|--------------------|--------|----------|
| VTV    | Vanguard Value ETF | 109.2% |          |

Sector 1: U.S. Large Cap Equity

- Relative-strength state: **FLAT** RS momentum (63d): **-2.5%** held: **3 / 81** members
- Theme: U.S. Large Cap Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| QUAL   | iShares MSCI USA Quality Fa... | 7.2%   | 3.7%   | FLAT   | 23        |
| QDF    | FlexShares Quality Dividend... | 7.0%   | 3.9%   | FLAT   | 23        |
| ITA    | iShares U.S. Aerospace & De... | 3.3%   | 1.7%   | FLAT   | 23        |

Strategy-1 replication basket (context):

| ticker | name                 | rep_wt | held_now |
|--------|----------------------|--------|----------|
| V00    | Vanguard S&P 500 ETF | 100.7% |          |

## Closed Trades

The strategy rebalances **monthly** – signals are computed daily, but positions only change at month-end (last trading day decision, execution next morning). The closed-trade ledger below is derived from the *engine's actual book*: each row is a position that the book opened and later closed, with hold duration measured from when the name first appeared in the weights to when it dropped out.

Closed positions: **404**. Mean hold: **43 days**. Median hold: **30 days**. Mean gross return: **2.1%** (absolute price over the held window).

Win rate: **57%**.

Table 26: Top 10 winning closed positions (engine-derived).

| ticker | entry      | exit       | hold_days | ret   |
|--------|------------|------------|-----------|-------|
| SOXX   | 2026-04-01 | 2026-06-30 | 90        | 81.3% |
| COPX   | 2025-12-01 | 2026-02-27 | 88        | 47.2% |
| CNXT   | 2025-08-01 | 2025-10-31 | 91        | 37.7% |
| EWT    | 2026-03-02 | 2026-05-29 | 88        | 34.7% |
| ARKW   | 2025-06-02 | 2025-08-29 | 88        | 27.2% |
| SMH    | 2024-05-01 | 2024-06-28 | 58        | 24.6% |
| COPX   | 2025-09-02 | 2025-10-31 | 59        | 21.9% |
| PTF    | 2024-10-01 | 2024-11-29 | 59        | 21.7% |
| XSD    | 2026-05-01 | 2026-06-30 | 60        | 20.8% |
| QTUM   | 2026-05-01 | 2026-06-30 | 60        | 20.4% |

Table 27: Top 10 losing closed positions (engine-derived).

| ticker | entry      | exit       | hold_days | ret    |
|--------|------------|------------|-----------|--------|
| URA    | 2026-02-02 | 2026-03-31 | 57        | -11.4% |
| GDX    | 2023-12-01 | 2024-01-31 | 61        | -11.4% |
| RFV    | 2023-02-01 | 2023-03-31 | 58        | -12.4% |
| RZV    | 2023-02-01 | 2023-03-31 | 58        | -12.9% |
| ECH    | 2026-02-02 | 2026-03-31 | 57        | -14.5% |
| GDXJ   | 2023-12-01 | 2024-01-31 | 61        | -14.5% |
| EZA    | 2026-03-02 | 2026-03-31 | 29        | -16.2% |
| PSI    | 2023-09-01 | 2023-10-31 | 60        | -19.2% |
| GDX    | 2026-03-02 | 2026-03-31 | 29        | -21.9% |
| GOAU   | 2026-03-02 | 2026-03-31 | 29        | -22.3% |

## ETF RS Leaders &amp; Laggards

The relative-strength analogue of strategy 1's Net Signal Strength: ETFs ranked by their **63-day relative-strength momentum** (ETF vs its own cluster PC1). Bullish list is restricted to ETFs in clusters currently in relative-LONG; bearish list is unrestricted.

Table 28: Top 20 most bullish ETFs by RS momentum (restricted to leading sectors).

| ticker | name                             | sector                  | rs_mom | held | etf_rs |
|--------|----------------------------------|-------------------------|--------|------|--------|
| GOAU   | US Global GO Gold and Preciou... | Gold                    | 14.9%  |      | FLAT   |
| GDX    | VanEck Gold Miners ETF           | Gold                    | 13.0%  |      | LONG   |
| EWS    | iShares MSCI Singapore ETF       | Developed ex-US Equity  | 12.6%  | Y    | LONG   |
| GDXJ   | VanEck Junior Gold Miners ETF    | Gold                    | 11.0%  |      | LONG   |
| EPOL   | iShares MSCI Poland ETF          | Developed ex-US Equity  | 6.9%   | Y    | LONG   |
| SIL    | Global X Silver Miners ETF       | Gold                    | 6.8%   |      | LONG   |
| NANR   | State Street SPDR S&P North A... | Materials               | 6.4%   |      | LONG   |
| EUFN   | iShares MSCI Europe Financial... | Developed ex-US Equity  | 6.2%   | Y    | FLAT   |
| COPX   | Global X Copper Miners ETF       | Materials               | 5.6%   |      | LONG   |
| ESPO   | VanEck Video Gaming and eSpor... | Developed ex-US Equity  | 5.4%   |      | FLAT   |
| IGE    | iShares North American Natura... | Materials               | 5.4%   |      | FLAT   |
| EWP    | iShares MSCI Spain ETF           | Developed ex-US Equity  | 5.2%   |      | FLAT   |
| PEY    | Invesco High Yield Equity Div... | U.S. Dividend Equity    | 5.2%   | Y    | LONG   |
| EWO    | iShares MSCI Austria ETF         | Developed ex-US Equity  | 5.0%   |      | LONG   |
| PWV    | Invesco Large Cap Value ETF      | U.S. Dividend Equity    | 4.1%   | Y    | LONG   |
| EZA    | iShares MSCI South Africa Ind... | Emerging Markets Equity | 4.0%   |      | FLAT   |
| SILJ   | Amplify Junior Silver Miners ... | Gold                    | 3.9%   |      | LONG   |
| EWT    | iShares MSCI Taiwan ETF          | Emerging Markets Equity | 3.4%   |      | LONG   |
| EWC    | iShares MSCI Canada Index Fund   | Developed ex-US Equity  | 3.3%   |      | LONG   |
| EWI    | iShares MSCI Italy ETF           | Developed ex-US Equity  | 3.2%   |      | FLAT   |

Table 29: Top 20 most bearish ETFs by RS momentum (full universe).

| ticker | name                             | sector                  | rs_mom | sector_rs |
|--------|----------------------------------|-------------------------|--------|-----------|
| NOBL   | ProShares S&P 500 Dividend Ar... | India Equity            | -63.0% | FLAT      |
| PTF    | Invesco Dorsey Wright Technol... | U.S. Large Cap Growth   | -21.9% | FLAT      |
| XSD    | State Street SPDR S&P Semicon... | U.S. Large Cap Growth   | -19.8% | FLAT      |
| ARKQ   | ARK Autonomous Technology & R... | Technology              | -19.6% | FLAT      |
| PRN    | Invesco Dorsey Wright Industr... | U.S. Large Cap Equity   | -18.5% | FLAT      |
| PSI    | Invesco Semiconductors ETF       | U.S. Large Cap Growth   | -14.4% | FLAT      |
| CNXT   | VanEck ChiNext Innovators ETF    | China Equity            | -14.2% | FLAT      |
| SLV    | iShares Silver Trust             | Gold                    | -13.9% | LONG      |
| SIVR   | abrdn Physical Silver Shares ... | Gold                    | -13.9% | LONG      |
| EWY    | iShares MSCI South Korea ETF     | Emerging Markets Equity | -12.7% | LONG      |

|      |                                  |                       |        |      |
|------|----------------------------------|-----------------------|--------|------|
| FTXL | First Trust Nasdaq Semiconduc... | U.S. Large Cap Growth | -12.2% | FLAT |
| BLOK | Amplify Blockchain Technology... | Technology            | -12.2% | FLAT |
| BOTZ | Global X Robotics & Artificia... | U.S. Large Cap Equity | -11.6% | FLAT |
| KGRN | KraneShares MSCI China Clean ... | China Equity          | -11.1% | FLAT |
| EIS  | iShares MSCI Israel ETF          | U.S. Large Cap Equity | -10.7% | FLAT |
| PKB  | Invesco Building & Constructi... | U.S. Large Cap Value  | -10.5% | FLAT |
| URA  | Global X Uranium ETF             | Materials             | -9.1%  | LONG |
| ROBO | ROBO Global Robotics and Auto... | U.S. Large Cap Equity | -9.1%  | FLAT |
| PPH  | VanEck Pharmaceutical ETF        | Healthcare            | -8.5%  | FLAT |
| IYZ  | iShares U.S. Telecommunicatio... | U.S. Large Cap Value  | -8.2%  | FLAT |

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## Watchlist

Table 30: Eligible leaders in leading sectors not currently held (next in line at next rebal).

| ticker | name                             | sector                  | rs_mom |
|--------|----------------------------------|-------------------------|--------|
| GDX    | VanEck Gold Miners ETF           | Gold                    | 13.0%  |
| GDXJ   | VanEck Junior Gold Miners ETF    | Gold                    | 11.0%  |
| SIL    | Global X Silver Miners ETF       | Gold                    | 6.8%   |
| NANR   | State Street SPDR S&P North A... | Materials               | 6.4%   |
| COPX   | Global X Copper Miners ETF       | Materials               | 5.6%   |
| EWO    | iShares MSCI Austria ETF         | Developed ex-US Equity  | 5.0%   |
| SILJ   | Amplify Junior Silver Miners ... | Gold                    | 3.9%   |
| EWT    | iShares MSCI Taiwan ETF          | Emerging Markets Equity | 3.4%   |
| EWG    | iShares MSCI Canada Index Fund   | Developed ex-US Equity  | 3.3%   |
| GNR    | State Street SPDR S&P Global ... | Materials               | 3.1%   |
| BBAX   | JPMorgan BetaBuilders Develop... | Developed ex-US Equity  | 2.7%   |
| REZ    | iShares Residential and Multi... | Real Estate             | 2.7%   |
| DBEF   | Xtrackers MSCI EAFE Hedged Eq... | Developed ex-US Equity  | 2.1%   |
| SDOG   | ALPS Sector Dividend Dogs ETF    | U.S. Dividend Equity    | 2.0%   |
| FEZ    | State Street SPDR EURO STOXX ... | Developed ex-US Equity  | 1.7%   |

## Run Parameters

| parameter                       | value        |
|---------------------------------|--------------|
| Holding mode                    | etf_topk     |
| Signal (short/buy/sell MA)      | 5 / 50 / 20  |
| Sell cushion k                  | 0.5          |
| Min hold (days)                 | 30           |
| Rearm blocked buys              | TRUE         |
| RS ranking window               | 63           |
| Leading clusters (n)            | 6            |
| Leaders per cluster (top_k)     | 3            |
| Weight scheme                   | inverse_vol  |
| Vol window (for inverse-vol)    | 60           |
| Per-name cap                    | 20%          |
| Relative entry filter (abs-mom) | 252d         |
| Absolute overlay (abs-mom)      | 252d         |
| Market gate                     | ON (200d MA) |
| Rebalance                       | monthly      |
| Cost (bps)                      | 10           |